

Code-Layout, Readability and Reusability

Date	30 September 2026
Team ID	SWTID-2026-6682
Project Name	Comic Craft AI
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Standard 4-space indentation followed across Python backend and JS frontend.
2	Proper File Structure	Files and folders are logically organized	Yes	Separated clean directory structure for frontend, backend, components, and static/assets.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Self-descriptive naming conventions used (e.g., comic_strip_prompt, generated_panel_image)
4	Function / Method Names	Functions are descriptively named	Yes	Action-oriented naming followed (e.g., generate_panel_image(), save_storyboard()).
5	Code Comments	Inline and block comments explain logic	Yes	Clear docstrings provided for API handlers and complex prompt formatting functions.
6	Modular Design	Code is split into reusable functions/modules	Yes	AI image generation, text processing, and canvas rendering logic are separated into standalone modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common utility functions used across components to avoid duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-except blocks implemented for external AI API calls and database transactions.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Image Generator Wrapper	Python / OpenAI / Stable Diffusion API	Comic panel generation & character portrait creation modules	High

2	Panel Canvas Grid	JavaScript / React / HTML5 Canvas	Comic page layout viewer & export preview component	High
3	Prompt Formatter	Python	Script generator and scene description generator	Medium
4	Speech Bubble Renderer	JavaScript / CSS	Custom canvas overlay for characters and captions	High
5	PDF / Image Exporter	Python / Pillow	Final comic book download and preview options	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project directory separation between frontend UI and AI backend endpoints.
Readability	5	Code follows standard PEP8 and JavaScript style guides with clean variable and function naming.
Reusability	4	Core generation modules and rendering scripts are highly decoupled and reusable.
Documentation / Comments	4	Comprehensive inline comments and function docstrings provided for all major modules.
Overall Score	4.5/5	Excellent software architecture with well-structured, maintainable, and clean code.